MLE with mis-specified model

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Warmup

Warmup: Poisson

Suppose $Y_1, ..., Y_n \overset{iid}{\sim} Poisson(\lambda)$.

$$\widehat{\lambda} = \overline{Y} \qquad \sqrt{n}(\widehat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda)$$

Want: variance-stabilizing transformation g such that asymptotic variance of $\sqrt{n}(g(\widehat{\lambda}) - g(\lambda))$ does not depend on λ

$$\lambda [g'(\lambda)]^2 = \text{constant}$$

$$\Rightarrow g'(\lambda) \propto \frac{1}{\sqrt{\lambda}}$$

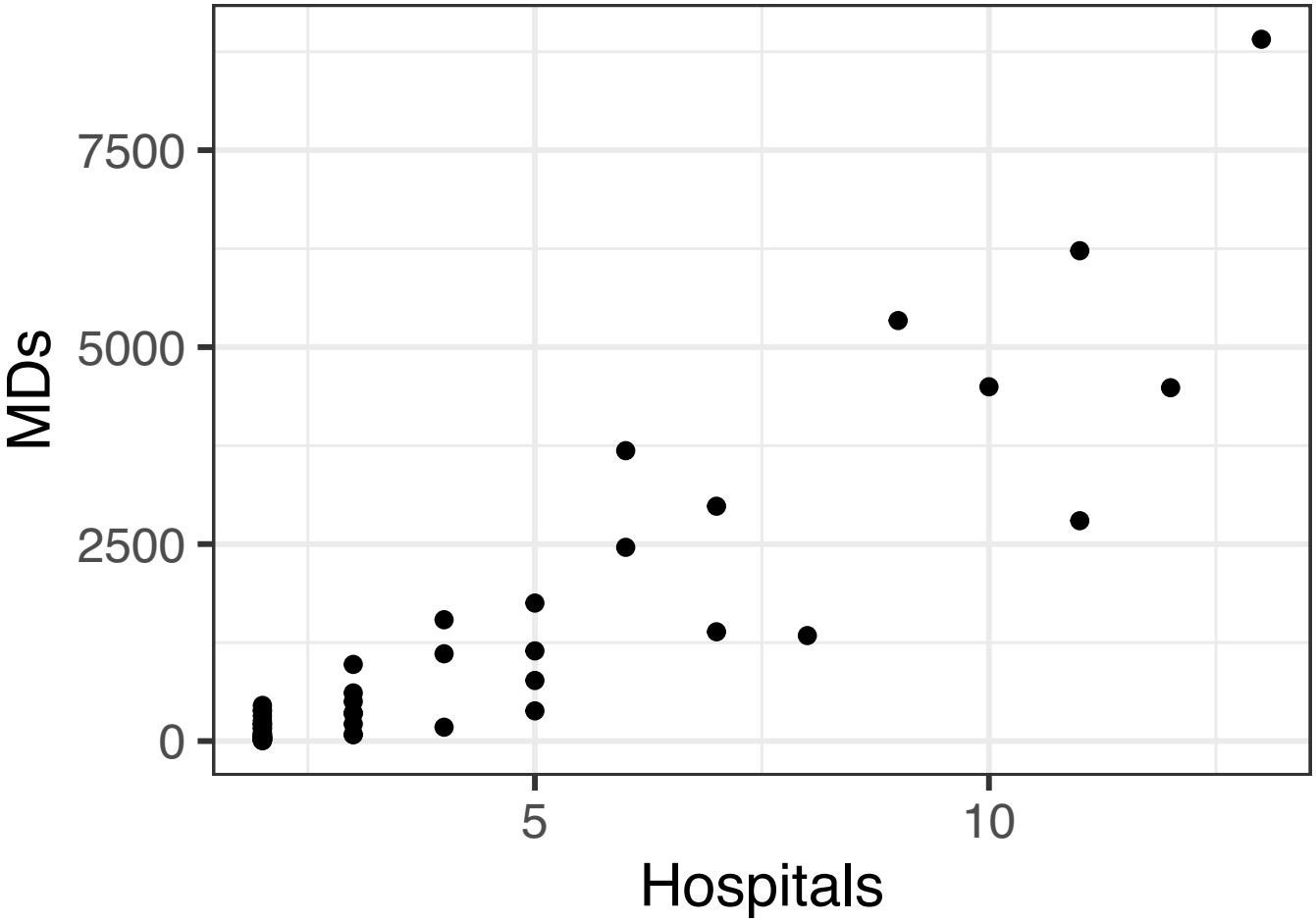
$$\Rightarrow g(\lambda) = \sqrt{\lambda} \qquad g'(\lambda) = \frac{1}{2}\lambda^{-\frac{1}{2}}$$

$$\Rightarrow \sqrt{n}(\sqrt{\hat{\lambda}} - \sqrt{\lambda}) \xrightarrow{d} N(0, \lambda(\tfrac{1}{2}\lambda^{-\frac{1}{2}})^2)$$

$$= N(0, \tfrac{1}{4})$$

Example: non-constant variance

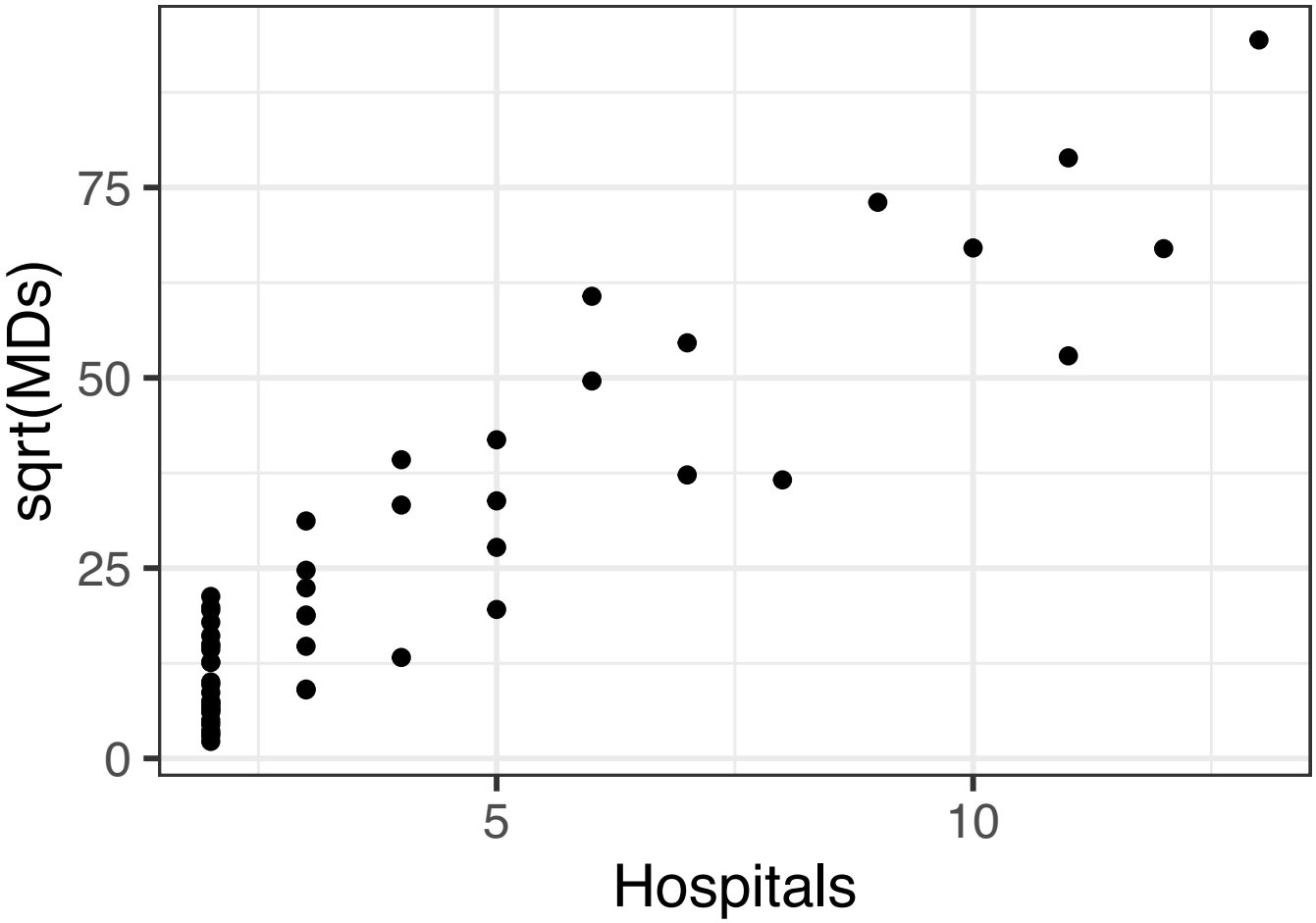
Example: Data on the number of hospitals and number of doctors (MDs) in US counties



Question: How do we adjust for non-constant variance in a linear model?

Example: non-constant variance

Example: Data on the number of hospitals and number of doctors (MDs) in US counties



Variance stablizing transformations

Variance stabilizing transformations are often used when there is a relationship between the *mean* and the *variance* – for example, transforming the response in a linear regression model. Different transformations address different mean-variance relationships. Some examples:

- If $\mathbb{E}[Y] = \mu$ and $Var(Y) = \mu$, then **square root** is a variance stabilizing transformation
- If $\mathbb{E}[Y] = \mu$ and $Var(Y) = \mu^2$, then **log** is a variance stablizing transformation

Why maximum likelihood estimators are nice

Let $\boldsymbol{\theta} \in \mathbb{R}^d$ be a parameter of interest for a distribution with probability function $f(y|\boldsymbol{\theta})$, and $\widehat{\boldsymbol{\theta}}$ the maximum likelihood estimator. Under regularity conditions, the MLE has some very nice properties:

- $\widehat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}$ (consistency)
- $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \underbrace{\mathcal{I}_1^{-1}(\boldsymbol{\theta})}_{\text{CRLB}})$ (asymptotic normality)

MLE is asymptotically unbiased

MLE is asymptotically efficient

(asymptotic variance is CRLB)

However: These results assume that we have correctly specified the data distribution $f(y|\boldsymbol{\theta})$! What happens if we were *wrong*?

Revisiting the asymptotic normality proof

Theorem: Let $Y_1, ..., Y_n \overset{iid}{\sim} f(y|\boldsymbol{\theta})$, and let $\widehat{\boldsymbol{\theta}}$ be the MLE of $\boldsymbol{\theta}$. Under regularity conditions,

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}_1^{-1}(\boldsymbol{\theta}))$$

Proof sketch: (in the case $\theta \in \mathbb{R}$)

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f(Y_i|\theta)$$

$$\sqrt{n}(\hat{\theta} - \theta) \approx \frac{\frac{1}{\sqrt{n}} \ell_n'(\theta)}{-\frac{1}{n} \ell_n''(\theta)}$$ (Taylor expansion)

$$\frac{1}{\sqrt{n}} \ell_n'(\theta) \xrightarrow{d} N\left(0, \operatorname{Var}\left(\frac{\partial}{\partial\theta} \log f(Y_i|\theta)\right)\right)$$ (CLT)

$$-\frac{1}{n} \ell_n''(\theta) \xrightarrow{p} -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(Y_i|\theta)\right]$$ (WLLN)

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{\operatorname{Var}\left(\frac{\partial}{\partial\theta} \log f(Y|\theta)\right)}{\left(-\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(Y|\theta)\right]\right)^2}\right)$$

under regularity,

$$\mathcal{I}_1(\theta) = \operatorname{Var}\left(\frac{\partial}{\partial\theta} \log f(Y|\theta)\right) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(Y|\theta)\right]$$

Asymptotics when the distribution is mis-specified?

Setup: Suppose that $Y_1, \ldots, Y_n \overset{iid}{\sim} g$ for some (unknown) distribution with probability function g. *However*, we instead assume that $Y_i \overset{iid}{\sim} f(y|\boldsymbol{\theta})$ for some specified distribution f with parameter $\boldsymbol{\theta}$.

Estimation: $\hat{\theta}$ still solves $\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(Y_i|\theta) = 0$

But what is $\hat{\theta}$ estimating??

Let θ_0 be value such that

$$E_g\left[\frac{\partial}{\partial \theta_0} \log f(Y_i|\theta_0)\right] = 0$$

expectation under the distribution g

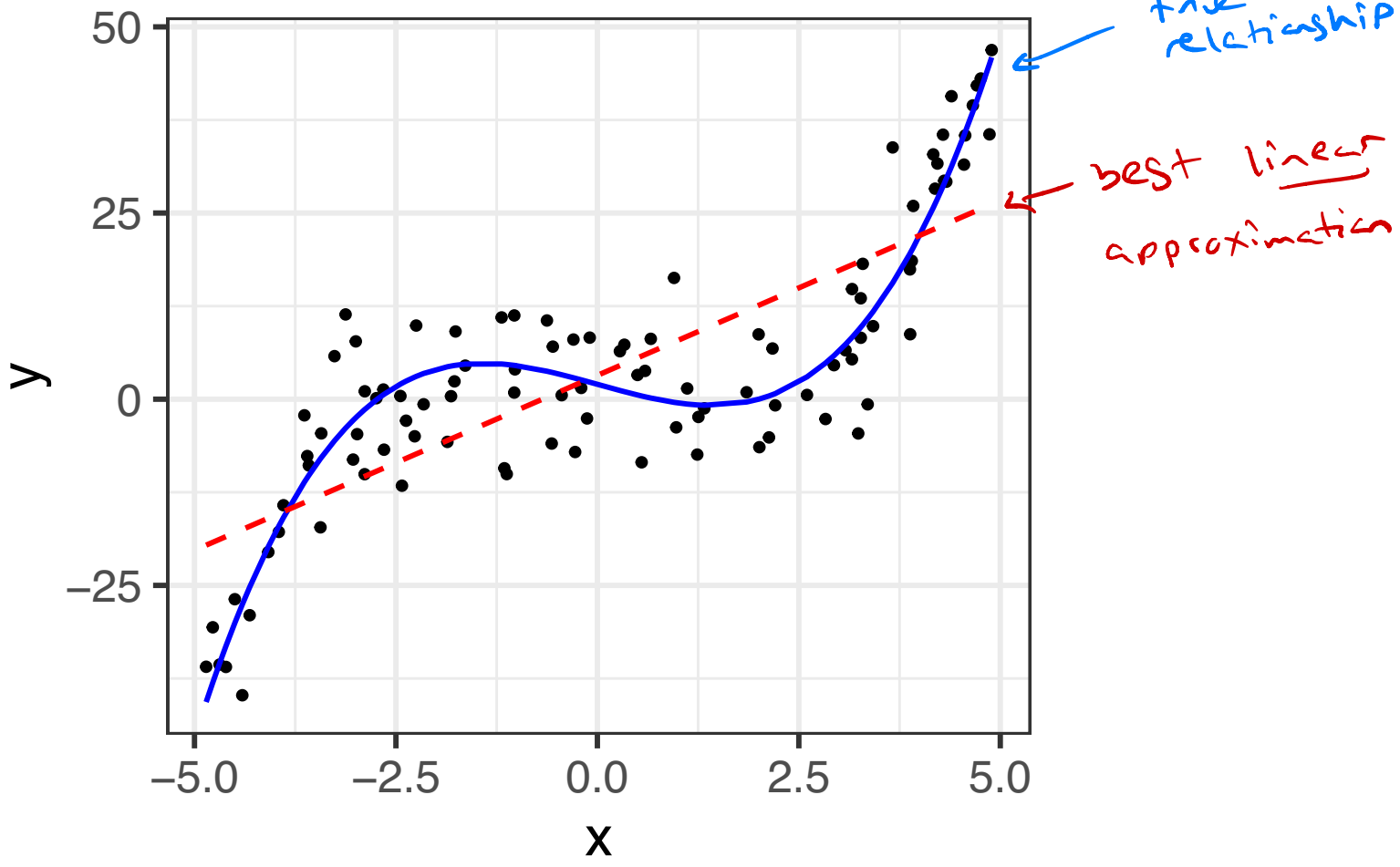
under regularity conditions, $\hat{\theta} \xrightarrow{p} \theta_0$

$f(y|\theta_0)$ is the best approximation to g for the chosen model f

Example from regression

Truth: (X_i, Y_i) with $Y_i|X_i \sim N(\gamma_0 + \gamma_1 X_i + \gamma_2 X_i^2 + \gamma_3 X_i^3, \sigma^2)$ ← g

Assumption: (X_i, Y_i) with $Y_i|X_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$ ← f



Asymptotics when the distribution is mis-specified?

Setup: Suppose that $Y_1, \ldots, Y_n \overset{iid}{\sim} g$ for some (unknown) distribution with probability function g. *However*, we instead assume that $Y_i \overset{iid}{\sim} f(y|\boldsymbol{\theta})$ for some specified distribution f with parameter $\boldsymbol{\theta}$.

$\hat{\theta}$ solves $\ell_n'(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \log f(Y_i|\theta) = 0$

θ_0 solves $E_g\left[\frac{\partial}{\partial\theta} \log f(Y|\theta)\right] = 0$

$\hat{\theta} \xrightarrow{p} \theta_0$ (under regularity)

As before: if $\theta \in \mathbb{R}$, $\sqrt{n}(\hat{\theta} - \theta_0) \approx \dfrac{\frac{1}{\sqrt{n}} \ell_n'(\theta_0)}{-\frac{1}{n} \ell_n''(\theta_0)}$

CLT: $\frac{1}{\sqrt{n}} \ell_n'(\theta_0) \xrightarrow{d} N\left(0, \mathrm{Var}_g\left(\frac{\partial}{\partial\theta_0} \log f(Y_i|\theta_0)\right)\right)$

WLLN: $-\frac{1}{n} \ell_n''(\theta_0) \xrightarrow{p} -E_g\left[\frac{\partial^2}{\partial\theta_0} \log f(Y_i|\theta_0)\right]$

$$\Rightarrow \sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \underbrace{\frac{\mathrm{Var}_g\left(\frac{\partial}{\partial\theta_0} \log f(Y_i|\theta_0)\right)}{\left(-E_g\left[\frac{\partial^2}{\partial\theta_0^2} \log f(Y_i|\theta_0)\right]\right)^2}}\right)$$

$$\left(-E_g\left[\frac{\partial^2}{\partial\theta_0^2} \log f(Y_i|\theta_0)\right]\right)^{-1} \mathrm{Var}_g\left(\frac{\partial}{\partial\theta_0} \log f(Y_i|\theta_0)\right)\left(-E_g\left[\frac{\partial^2}{\partial\theta_0^2} \log f(Y_i|\theta_0)\right]\right)^{-1}$$

Asymptotics when the distribution is mis-specified?

Setup: Suppose that $Y_1, ..., Y_n \overset{iid}{\sim} g$ for some (unknown) distribution with probability function g. *However*, we instead assume that $Y_i \overset{iid}{\sim} f(y|\boldsymbol{\theta})$ for some specified distribution f with parameter $\boldsymbol{\theta}$.

- Let $\boldsymbol{\theta}_0$ solve $\mathbb{E}_g\left[\frac{\partial}{\partial\boldsymbol{\theta}}\log f(Y|\boldsymbol{\theta})\right] = \mathbf{0}$.
- Let $\widehat{\boldsymbol{\theta}}$ solve $\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}}\log f(Y_i|\boldsymbol{\theta}) = \mathbf{0}$

Key results: Under regularity conditions,

- $\widehat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$
- $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S}(\boldsymbol{\theta}_0))$ where

$$\mathbf{S}(\boldsymbol{\theta}_0) = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$$

$$\mathbf{A} = -\mathbb{E}_g\left[\frac{\partial^2}{\partial\boldsymbol{\theta}_0^2}\log f(Y_i|\boldsymbol{\theta}_0)\right] \qquad \mathbf{B} = Var_g\left(\frac{\partial}{\partial\boldsymbol{\theta}_0}\log f(Y_i|\boldsymbol{\theta}_0)\right)$$